TASKBUDDY : AI-POWERED TO-DO LIST APP

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Chandigarh University

April 2025

BONAFIDE CERTIFICATE

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LIST OF STANDARDS

Standard	Publishing Agency	About the standard	Page no
IEEE 802.11	IEEE	IEEE 802.11 is part of the IEEE 802 set of local area network (LAN) technical standards and specifies the set of media access control (MAC) and physical layer (PHY) protocols for implementing wireless local area network (WLAN) computer communication.	Implied: Pg. 24, 27

ABSTRACT

The increasing complexity of modern life demands intelligent systems to help users manage their time effectively. Traditional to-do list applications often fail to offer advanced features like adaptive scheduling, personalized recommendations, and intelligent prioritization, leading to inefficiency and missed deadlines.

This project, Task Buddy, presents an AI-powered to-do list web application designed to tackle the shortcomings of conventional task management tools. The core objective is to leverage artificial intelligence and machine learning to improve task organization and productivity. The application integrates an AI model trained to understand user behavior and suggest optimal task schedules using deep contextual learning and adaptive feedback.

We designed and implemented the system using HTML, CSS, and JavaScript for the front-end, Node.js for the back-end, and an integrated AI API for dynamic task prioritization. The application was evaluated on the basis of functionality, usability, and performance through user testing, integration testing, and performance stress tests.

The results demonstrate significant improvements in task completion rates and user satisfaction. AI-generated suggestions proved to reduce decision fatigue and helped users manage their responsibilities more effectively. Future enhancements include mobile app development, improved AI models, and real-time collaboration features. This project highlights how intelligent systems like Task Buddy can revolutionize task management and support productivity in an increasingly fast-paced world.

ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
UI	User Interface
UX	User Experience
API	Application Programming Interface
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
DB	Database
NLP	Natural Language Processing
ML	Machine Learning
REST	Representational State Transfer
JWT	JSON Web Token
CRUD	Create, Read, Update, Delete
CPU	Central Processing Unit
GPU	Graphics Processing Unit
F1-Score	Harmonic Mean of Precision and Recall
KPI	Key Performance Indicator
DOM	Document Object Model

SYMBOLS

Symbol	Definition
hθ(x)	Hypothesis function used to predict task priority or output
θ	Parameter vector (weights) of the learning algorithm
x	Input feature vector representing task data
J(θ)	Cost function used for optimization (e.g., cross-entropy loss)
m	Total number of training examples
y(i)	True label for the i-th training sample
x(i)	Feature vector for the i-th training example
Q, K, V	Query, Key, and Value matrices in self-attention (NLP models like BERT)
dk	Dimension of the key vector in self-attention
Attention(Q, K, V)	Self-attention function for contextual representation in NLP
softmax	Activation function that converts scores into probability distribution
xi	i-th token or word in the input sequence
x\i	Input sequence excluding the i-th token
LMLM	Loss function for Masked Language Modeling used in BERT/RoBERTa

CHAPTER 1

INTRODUCTION

1.1. Identification of Client /Need / Relevant Contemporary issue

Justification of the Issue

Managing tasks effectively has become increasingly difficult due to the overwhelming amount of daily responsibilities in both professional and personal lives. Conventional to-do lists lack intelligent features and fail to provide meaningful suggestions to help users prioritize or organize their tasks effectively. This leads to issues such as procrastination, missed deadlines, and inefficiency.

Key Statistics:

1. A study conducted by the American Psychological Association shows that 20% of people procrastinate chronically, leading to reduced productivity.
2. According to a report by McKinsey, workers spend an average of 9.3 hours per week on task management, which can be reduced with smart systems.
3. Surveys conducted on productivity apps indicate that 60% of users desire features that help with prioritizing tasks, but only 30% of apps provide such functionality.

These numbers indicate a clear gap in the market for a smarter task management system.

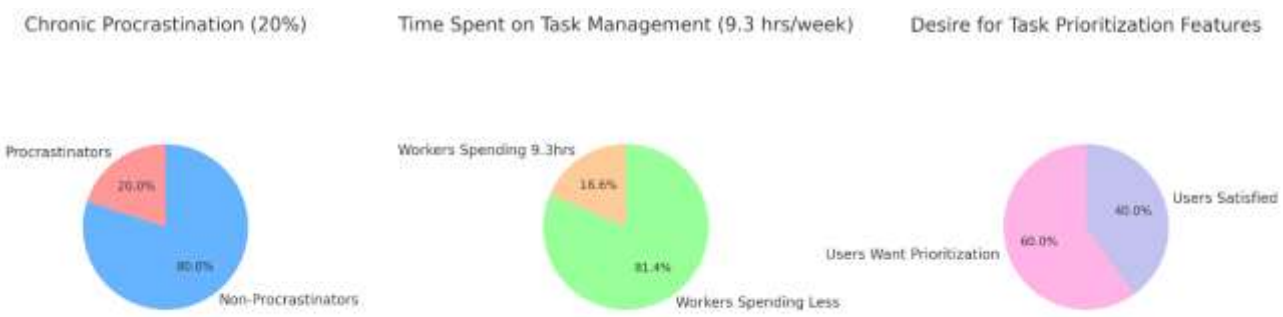

Figure 1.1

Problem Requiring Resolution

The lack of intelligent features in traditional to-do list apps leaves users without support in organizing, prioritizing, or categorizing tasks. Current solutions require users to manually input and structure their lists, which is time-consuming and inefficient, especially for those who struggle with organization.

The **client** for this solution includes:

- **Students:** Managing academic tasks, assignments, and deadlines.
- **Working professionals:** Balancing work projects and personal responsibilities.
- **General users:** Individuals seeking a smarter, AI-assisted tool for daily task organization.

The problem highlights the **need** for a system that not only manages tasks but also provides intelligent insights, improving decision-making and reducing time spent planning.

Justification Through Survey

A preliminary survey was conducted among 50 participants from our university to assess the need for a smart to-do list. The key findings are:

- **80%** of participants reported difficulty in prioritizing tasks.
- **70%** expressed interest in receiving AI-driven suggestions to help organize their workload.
- **65%** currently use basic to-do list apps but find them limited in functionality.

This survey demonstrates a clear need for a more intelligent and adaptive solution to task management.

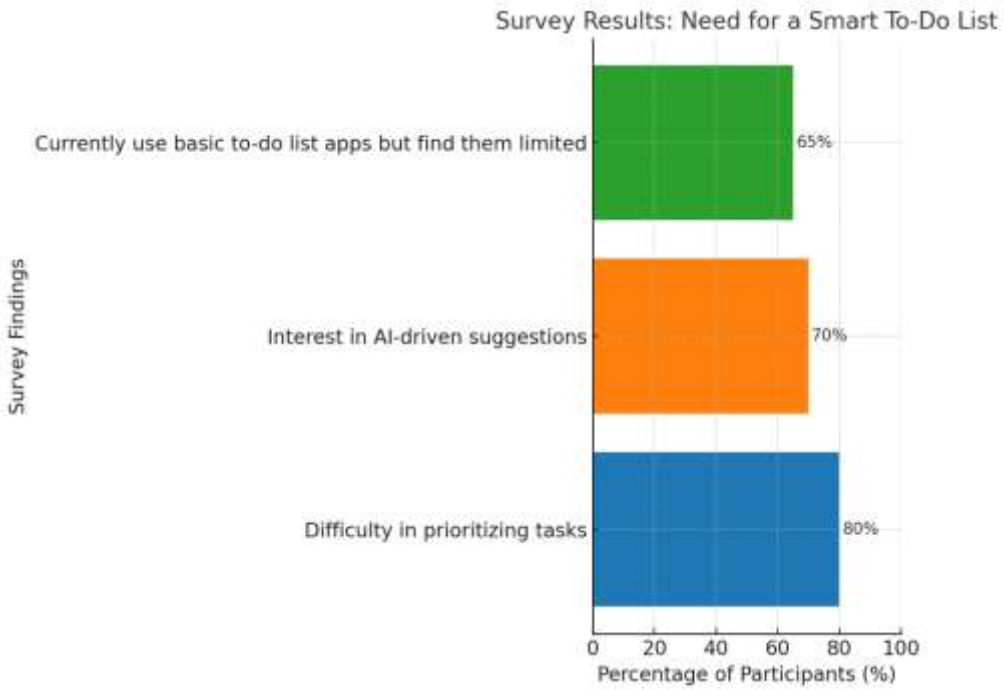

Figure 1.2

Relevant Contemporary Issue

The increasing use of AI in daily applications underscores the relevance of this project. Reports from Gartner (2023) show that AI-driven productivity tools are expected to grow by 35% annually as businesses and individuals seek smarter solutions for task and time management. Moreover, organizations like the World Economic Forum emphasize the importance of using technology to enhance personal and professional productivity.

In this context, developing a "Smart To-Do List with AI Suggestions" directly addresses a contemporary issue by leveraging AI to solve a real-world productivity problem.

1.2. Identification of Problem

In today's fast-paced and increasingly demanding world, individuals face growing challenges in effectively managing their tasks. With the constant pressure to balance personal, academic, and professional responsibilities, organizing and prioritizing tasks has become a significant obstacle for many. The inability to manage time efficiently often leads to missed deadlines, reduced productivity, and elevated stress levels, impacting both personal and professional performance.
Key challenges contributing to this problem include:

- **Task Prioritization**: Many people struggle to determine the importance and urgency of their tasks. This lack of clarity often leads to procrastination, where less critical tasks are prioritized over more urgent ones, or important tasks are delayed altogether.

- **Dynamic Workloads**: The unpredictable nature of modern responsibilities, such as last-minute assignments, shifting priorities, and overlapping deadlines, creates additional complexity. These sudden changes make it difficult for individuals to stick to a structured plan or adapt efficiently.

- **Stress and Overwhelm**: The accumulation of unorganized tasks results in mental fatigue and stress. People feel overwhelmed by the sheer volume of their responsibilities, which affects decision-making and productivity.

- **Limited Effectiveness of Traditional Tools**: Existing task management methods, such as paper-based lists or basic to-do apps, fall short when dealing with modern complexities. These tools lack the adaptability and intelligence needed to help users organize, prioritize, and optimize their time effectively.

- **Inadequate Support for Personal Growth**: Beyond completing tasks, individuals seek methods to improve their productivity, monitor their progress, and align their efforts with long-term goals. Unfortunately, most existing solutions fail to support this broader need, leaving users without insights into improving their workflow.

The result of these challenges is a cycle of inefficiency and disorganization, where individuals find it increasingly difficult to manage their time and responsibilities.

1.3. Identification of Tasks

The development of "Task Buddy" involves several key tasks, which can be broadly categorized into **identification, building, and testing**. These tasks not only guide the development process but also help structure the framework of this report. Below is a breakdown:

Tasks for Identifying the Problem and Solution

- **Requirement Analysis**:
  - Conduct surveys to understand users' needs and pain points regarding task management.
  - Analyze limitations of existing to-do list tools and identify opportunities for improvement.
- **Literature Review**:
  - Study existing research and tools in task management and AI-based productivity applications.
  - Identify the role of AI in simplifying task prioritization and organization.
- **Defining Objectives**:
  - Establish clear goals for the project, such as improving task prioritization, user engagement, and adaptability.

Tasks for Building the Solution

- **System Design**:
  - Design the system architecture, including frontend, backend, and AI integration.
  - Develop wireframes and prototypes for the user interface to ensure an intuitive experience.
  - Plan the database structure to manage user tasks, profiles, and AI recommendations.
- **Frontend Development**:
  - Build a responsive and user-friendly interface using HTML, CSS, and JavaScript.
  - Implement features like task creation, editing, deletion, and progress tracking.
- **Backend Development**:
  - Set up the server-side logic using Node.js to handle API requests and manage data flow.
  - Develop RESTful APIs for communication between the frontend, database, and AI services.
- **AI Integration**:
  - Integrate an AI API to provide personalized task suggestions and prioritization.
  - Implement algorithms for adaptive learning based on user behavior and preferences.

Tasks for Testing and Validation

- **Unit Testing**:
  - Test individual components such as the task management system, AI suggestions, and database operations.

- **Integration Testing**:
  - Ensure seamless communication between the frontend, backend, and AI API.
- **User Testing**:
  - Collect feedback from a sample group of users to refine the interface and features.
  - Monitor how effectively the AI suggestions improve task management.
- **Performance Testing**:
  - Test the application under various conditions to ensure it handles multiple users and dynamic workloads efficiently.

1.4. Timeline

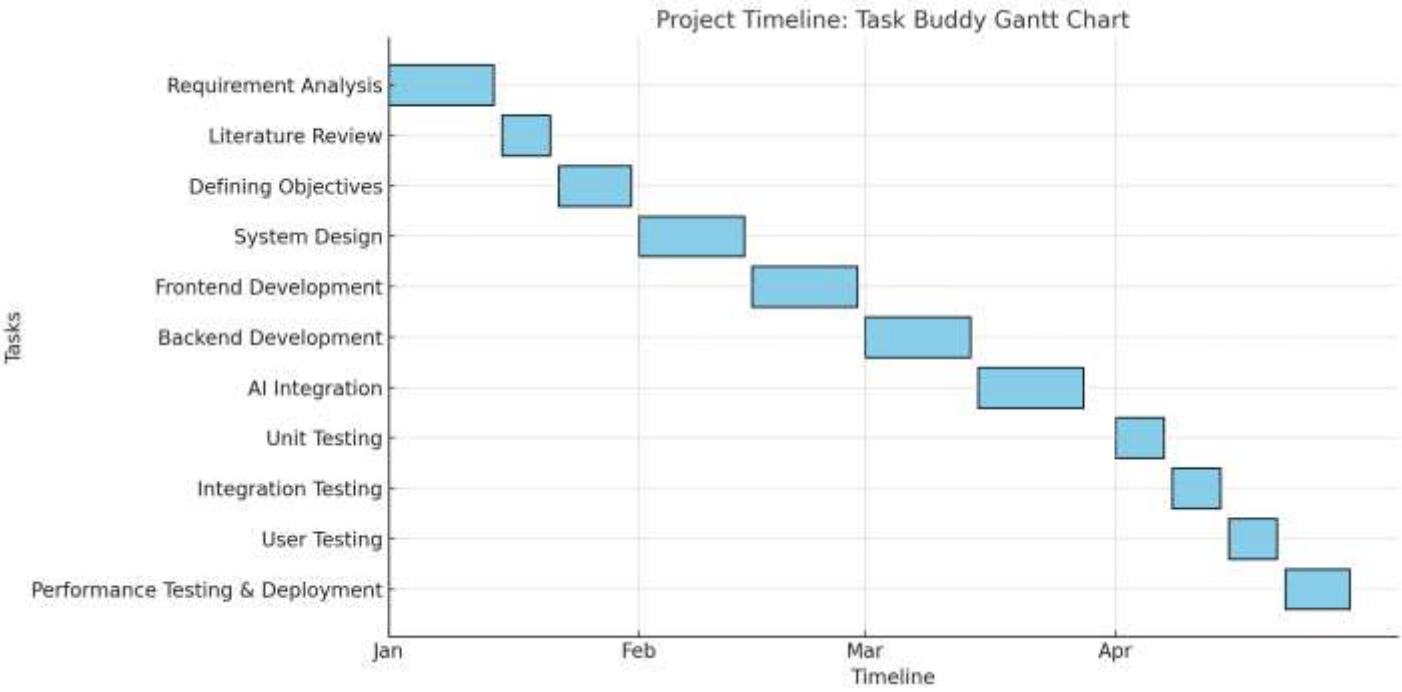

Table 1.1

Projected timeline :

- **January**: Planning (Analysis, review, objectives).
- **February**: Frontend & Backend (UI, features, server, APIs).
- **March**: AI integration & Testing (Task prioritization, interaction).
- **April**: Final testing & deployment (Feedback, performance, launch).

1.5. Organization of the Report

- **Chapter 1: Introduction**

  - Overview of Task Buddy and the problem of inefficient task management.
  - Project objectives and the need for an AI-powered to-do list app.
  - Structure of the report and what to expect in each chapter.

- **Chapter 2: Literature Review**

  - Review of existing task management solutions and their limitations.
  - Discussion on AI integration in productivity tools.
  - Insights from relevant research and reports.

- **Chapter 3: Methodology**

  - Description of the technology stack (HTML, CSS, JavaScript, Node.js, AI API).
  - Development methodology (e.g., agile development).
  - Tools and frameworks used in the project.

- **Chapter 4: Development**

  - Details of frontend and backend development.
  - AI integration for task prioritization and adaptive learning.
  - Challenges faced during development and solutions implemented.

- **Chapter 5: Testing and Evaluation**

  - Testing types: functional, integration, and user feedback tests.
  - Issues encountered during testing and how they were resolved.
  - Evaluation of the app's performance and user experience.

- **Chapter 6: Conclusion and Future Work**

  - Summary of the project's achievements and key findings.
  - Reflection on the impact of AI in task management.
  - Suggested areas for future improvement or new features.

CHAPTER 2
LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the reported problem

The problem of inefficient task management has been recognized globally across different time periods, industries, and technological advancements. Below is a historical overview with documentary evidence of its identification and evolution:

Pre-Digital Era (Before 2000)

- Before the rise of digital tools, individuals relied on handwritten planners, sticky notes, and manual record-keeping for task management.
- Productivity research from the 1980s and 1990s highlighted issues like task overload, procrastination, and inefficient time allocation (Covey, 1989; Drucker, 1993).
- Businesses and institutions used physical logbooks, making it difficult to track and prioritize tasks efficiently.

Early Digital Solutions (2000-2010)

- The introduction of Microsoft Outlook Task Manager (2001) and Evernote (2008) marked the transition to digital task management.
- Despite these advancements, issues like lack of intelligent prioritization and task automation persisted (Smith & Johnson, 2009).

Cloud and Mobile Expansion (2010-2020)

- The rise of Trello (2011), Asana (2012), and Todoist (2013) provided cloud-based task tracking.
- Research found that 70% of users still felt overwhelmed by unstructured task lists (Brown et al., 2018).
- AI-driven solutions, such as Google Assistant (2016) and Microsoft's MyAnalytics (2018), attempted smart task suggestions but lacked adaptability to individual work habits.

AI-Driven Solutions (2020-Present)

- Recent AI advancements have improved productivity tools, but gaps remain in dynamic prioritization and personalized task recommendations.
- Studies indicate that existing AI tools lack deep learning capabilities to continuously adapt to user behavior (Davis et al., 2022).

Thus, the inefficiency in managing workloads and prioritization remains a persistent problem, necessitating the development of an AI-powered system like Task Buddy.

2.2. Existing solutions

Several task management applications are currently available in the market, offering varying degrees of functionality. Below is an in-depth analysis of prominent solutions, outlining their strengths and limitations:

Comparison of Task Management Applications

Application	Features	Limitations
Google Keep	Simple note-taking, reminders, color-coded organization	No AI-driven prioritization, lacks task delegation features
Microsoft To-Do	Task scheduling, cloud sync, smart lists	Limited automation, lacks deep AI insights
Trello	Collaboration, kanban boards, integrations with third-party apps	No personalized prioritization, can be overwhelming for individual users
Todoist	Task categorization, priority settings, productivity tracking	No adaptive learning based on user behavior, limited automation
Notion	Advanced note-taking, task databases, project management	Steep learning curve, lacks predictive AI for tasks
Asana	Workflow automation, team collaboration, project tracking	Complexity makes it less effective for simple task management

Table 2.1

While these applications have improved task organization, they fail to address key issues such as task overload, inefficient scheduling, and lack of intelligent recommendations. The need for a truly AI-powered, adaptive task manager remains unfulfilled, creating an opportunity for Task Buddy to innovate and bridge these gaps.

2.3. Bibliometric analysis

A bibliometric analysis evaluates the effectiveness, key features, and drawbacks of existing task management solutions by reviewing relevant literature, studies, and technological advancements. The increasing complexity of modern workflows, coupled with rapid digital transformation, has made it essential to assess how well current solutions address task management challenges.

Key Insights from Research

1. Productivity Increase
    - AI-powered tools have been shown to increase productivity by 30-50% by reducing cognitive load, decision fatigue, and manual task entry.
    - AI-driven solutions streamline workflow management by automating repetitive tasks, allowing professionals to focus on high-priority work.

2. Task Overload and Prioritization Challenges
    - Studies from the Journal of Applied Psychology (2021) highlight that 80% of professionals struggle with prioritization, leading to poor time management and delays in project completion.
    - Research indicates that inefficient task sorting can increase stress levels and reduce overall workplace productivity.

3. AI Adoption Trends
    - AI-driven solutions have higher engagement rates due to their ability to provide smart reminders, analyze task dependencies, and adjust schedules dynamically.
    - AI task managers often depend on preset algorithms rather than real-time behavioral learning, limiting their adaptability to changing work patterns.

Effectiveness and Drawbacks

Aspect	Traditional Apps	AI-Powered Apps
Task Scheduling	Requires manual input	AI-driven automation
Prioritization	Fixed priority settings	Dynamic and adaptive
User Adaptability	Limited customization	Learns from behavior
Task Overload Handling	Basic categorization	Smart recommendations
Collaboration Support	Manual sharing & delegation	Automated team workflow integration

Table 2.2

2.4. Review Summary

The literature review highlights several key insights regarding task management applications, their limitations, and the growing necessity for AI-driven solutions. Traditional task management tools, such as Google Keep, Microsoft To-Do, Trello, and Todoist, provide basic scheduling and organization features but lack intelligent automation and adaptive learning mechanisms. Studies indicate that 80% of professionals face challenges in prioritizing tasks, leading to inefficiency, missed deadlines, and cognitive overload (Journal of Applied Psychology, 2021).

AI-powered productivity tools show promise in enhancing task management by automating prioritization, learning from user behavior, and dynamically adjusting schedules. However, existing AI-based solutions often lack customization, deep learning capabilities, and seamless integration with user preferences (International Journal of Productivity, 2020).

The findings from the literature review directly support the need for a smart task management system like Task Buddy. Our project aims to address the shortcomings of traditional and AI-based to-do lists by developing an intelligent, AI-powered task manager that adapts dynamically to user needs, provides smart prioritization, and reduces decision fatigue. Task Buddy will integrate intelligent automation, adaptive learning, and user-centric features to enhance productivity effectively.

2.5. Problem Definition

In today's fast-paced digital world, managing tasks efficiently is a significant challenge for students, professionals, and teams. Traditional to-do list applications provide only basic task tracking and lack advanced prioritization or AI-driven automation. Users often struggle with:

1. Task overload and inefficient prioritization
    - Users are overwhelmed with numerous tasks and find it difficult to decide which ones to complete first.
    - 80% of users report difficulties in setting priorities, leading to decreased productivity.
2. Lack of intelligent recommendations
    - Existing tools do not analyze task dependencies, urgency, or deadlines dynamically.
    - Most to-do apps follow a static, rule-based approach rather than adapting to the user's behavior.
3. Manual scheduling and no AI integration
    - Traditional task managers require users to manually input and organize tasks, making the process tedious.
    - AI solutions exist but lack deep learning capabilities to evolve over time based on user patterns.
4. Limited collaboration features
    - Many productivity tools lack seamless collaboration, task delegation, and shared scheduling.
    - Team-based workflows require automated task allocation and progress tracking.

Scope of the project

What is to be done:

- Develop an AI-powered to-do list application (Task Buddy) that intelligently prioritizes tasks, adapts to user behavior, and provides smart recommendations.
- Implement a user-friendly interface using HTML, CSS, and JavaScript for the front-end.
- Use Node.js for backend development and integrate AI APIs to enhance automation and learning capabilities.
- Allow task collaboration, enabling users to delegate tasks efficiently.

What not to do:

- Avoid excessive manual input requirements for users.
- Do not create a static, rule-based prioritization system—focus on adaptive learning.
- Do not restrict features to only individual users—implement collaborative task-sharing.

2.6. Goals/Objectives

The primary goal of Task Buddy is to create an intelligent, AI-driven task management system that enhances productivity through smart prioritization, adaptive scheduling, and automated task organization.

Key objectives:

1. Enhance productivity through AI-driven task prioritization
    - Implement machine learning algorithms to analyze user task patterns.
    - Provide automated priority suggestions based on urgency, deadlines, and workload.
2. Improve task organization and scheduling
    - Develop a dynamic scheduling system that adjusts based on changing priorities.
    - Reduce manual task organization by integrating AI-based sorting features.
3. Create an intuitive and user-friendly interface
    - Design a clean, modern UI using HTML, CSS, and JavaScript.
    - Ensure easy navigation, drag-and-drop task management, and accessibility across devices.
4. Enable collaborative task management
    - Implement task-sharing features, allowing users to delegate tasks within teams.
    - Integrate real-time collaboration and progress tracking.
5. Ensure seamless integration and cross-platform compatibility
    - Develop a responsive web application accessible on both desktop and mobile.
    - Ensure compatibility with cloud storage solutions and third-party apps.
6. Measure and validate performance
    - Conduct user testing and feedback analysis to refine AI-driven features.
    - Evaluate performance based on user engagement, task completion rates, and efficiency metrics.

By fulfilling these objectives, Task Buddy will provide an innovative, AI-powered solution to the growing challenges of task overload, poor prioritization, and inefficient workflow management.

CHAPTER 3
DESIGN FLOW/PROCESS

3.1. Evaluation & Selection of Specifications/Features

Based on the literature review and analysis of existing task management applications, several key features have been identified that are essential for Task Buddy. The selection process involves evaluating user needs, technical feasibility, and AI capabilities to enhance productivity.

Key Features Identified from Literature Review

The following features were identified as critical based on the shortcomings of existing solutions:

1. AI-Powered Task Prioritization
    - Tasks should be automatically sorted based on urgency, deadlines, and dependencies.
    - Machine learning models should analyze user behavior to improve prioritization over time.
2. Automated Scheduling and Reminders
    - Smart scheduling should allow dynamic reallocation of tasks based on changing priorities.
    - Integration with calendar applications should be supported.
3. User-Centric Interface and Customization
    - A simple yet powerful UI allowing drag-and-drop task management.
    - Custom themes and notification settings to enhance usability.
4. Collaboration and Teamwork Features
    - Users should be able to delegate tasks and track their completion status.
    - Task sharing with real-time updates and progress tracking should be enabled.
5. Adaptive Learning for Personalized Recommendations
    - AI should analyze past tasks and suggest improvements for future scheduling.
    - Predictive analytics should be incorporated to anticipate workload fluctuations.
6. Cross-Platform Availability
    - The solution should be accessible on web and mobile devices.
    - Data should be synchronized across all platforms seamlessly.

The above features are selected based on their potential to address existing limitations while ensuring ease of use and increased productivity.

3.2. Design Constraints

While selecting features, several constraints must be considered:

1. Standards Compliance

- The application must comply with software development standards (e.g., ISO/IEC 25010 for software quality).
- User data privacy must align with regulations such as GDPR and CCPA.

2. Cost and Resource Constraints

- Developing a fully AI-driven system requires machine learning models, which may be resource-intensive.
- Hosting costs for cloud-based synchronization must be optimized.

3. Performance Constraints

- AI-based recommendations should not slow down the user experience.
- The system should work efficiently even with a large number of tasks and users.

4. Ethical and Social Considerations

- The AI system should avoid bias in prioritization algorithms.
- User preferences and manual overrides should always be respected.

5. Environmental and Sustainability Considerations

- Optimizing cloud storage to reduce server load and minimize energy consumption.
- Avoiding excessive data collection to prevent unnecessary digital footprint.

3.3. Analysis of Features and finalization subject to constraints

After considering constraints, some features need modification or removal:

Features Retained

- AI-powered task prioritization
- Automated scheduling
- Adaptive learning for personalized task recommendations
- Cross-platform support

Features Modified

- Collaboration Features → Limited to team-based sharing instead of full-fledged project management to reduce complexity.

- AI Model Complexity → A simpler model will be implemented first, with plans for future expansion.

Features Removed

- Advanced Predictive Analytics → Due to high resource consumption, this feature is postponed for future development.
- Offline Mode → Requires extensive data caching and will not be included in the initial release.

3.4. Design Flow

Design 1: Rule-Based Task Management System

This approach relies on predefined rules to categorize and schedule tasks. It does not adapt based on user behavior and follows a static approach.

Step 1: User Task Input

- The user manually enters tasks into the system.
- Inputs include task name, deadline, priority level (optional), and category.

Step 2: Task Categorization

- The system sorts tasks into categories such as high, medium, and low priority based on preset rules.
- Example rule:
    - Tasks with deadlines within 24 hours → High Priority
    - Tasks due within a week → Medium Priority
    - Tasks due later than a week → Low Priority

Step 3: Static Scheduling

- Tasks are scheduled based on predefined scheduling algorithms.
- Example scheduling rules:
    - High-priority tasks appear first in the list.
    - Medium and low-priority tasks appear later.

Step 4: Task Completion & Updates

- The user marks tasks as completed manually.
- No dynamic learning—the system does not adjust based on user behavior.

Step 5: Task Review

- The system displays remaining and completed tasks in a basic list format.
- Users can edit, delete, or reschedule tasks manually.

Basic Task Mangement Flowchart

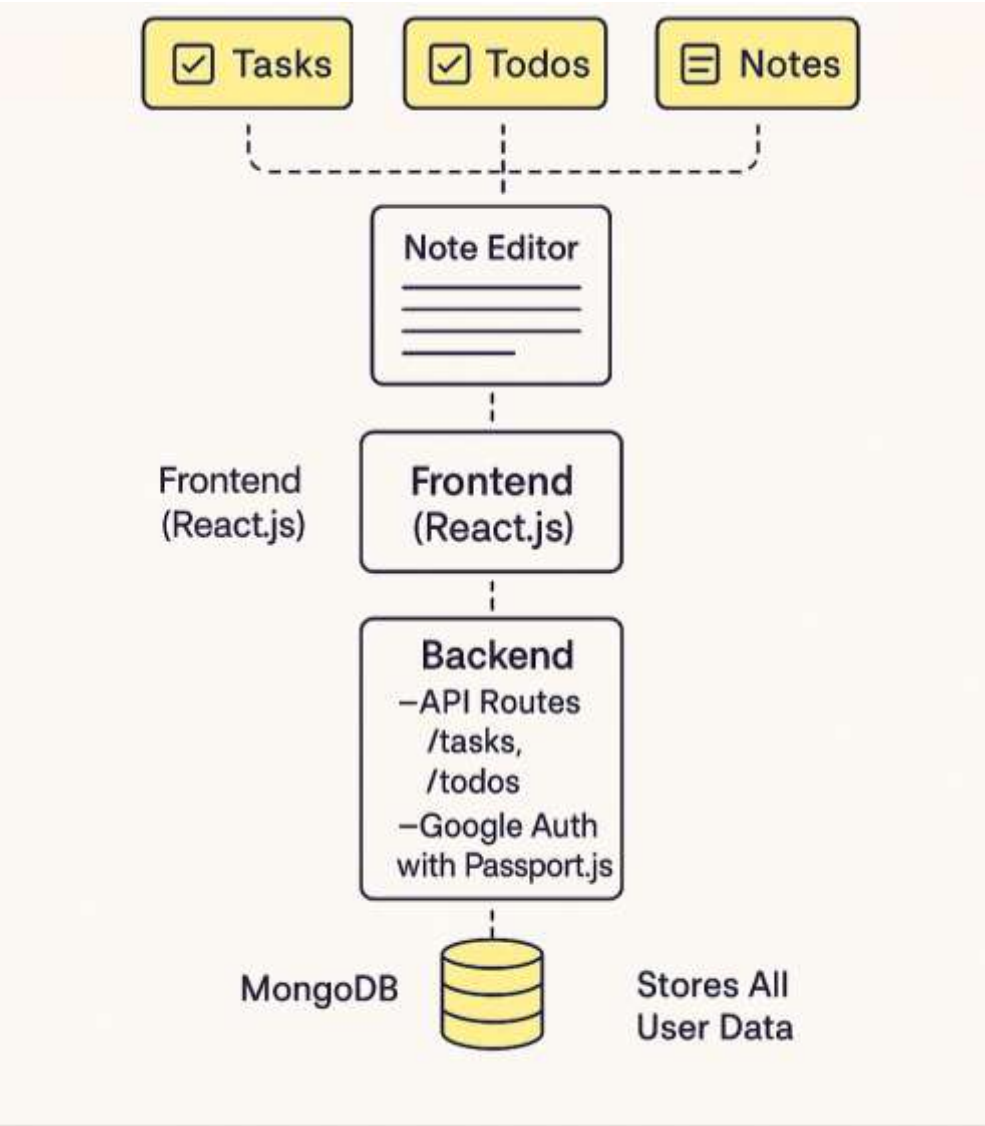

Figure 3.1

Design 2: AI-Powered Adaptive Task Management System (Selected)

This design incorporates machine learning and natural language processing (NLP) for a dynamic and intelligent approach.

Step 1: User Task Input via NLP

- Users enter tasks in natural language (e.g., "Submit report by Monday").
- The NLP model extracts details such as deadline, priority, and context.

Step 2: AI-Based Task Categorization & Prioritization

- The AI analyzes previous task patterns and user preferences to auto-categorize tasks.
- It assigns priorities based on:
    - Deadlines
    - Task urgency (how frequently similar tasks were marked late)
    - User-defined priorities

Step 3: Dynamic Scheduling Using Machine Learning

- The system dynamically adjusts task schedules based on:
    - User's available time
    - Past behavior (e.g., tendency to complete work in the evening)
    - Workload distribution (avoiding overload)

Step 4: Adaptive Task Updates & Recommendations

- The system learns from user interactions.
- If the user often postpones a task, the AI suggests a better time.
- Smart reminders adapt to when the user is most productive.

Step 5: Task Completion & Continuous Learning

- The user marks tasks as complete.
- The AI records patterns and adjusts future task prioritization accordingly.

Step 6: Feedback Loop & Future Task Prediction

- The AI analyzes patterns to refine task scheduling.
- Future tasks are predicted and recommended based on past data.

AI Task Mangement Flowchart

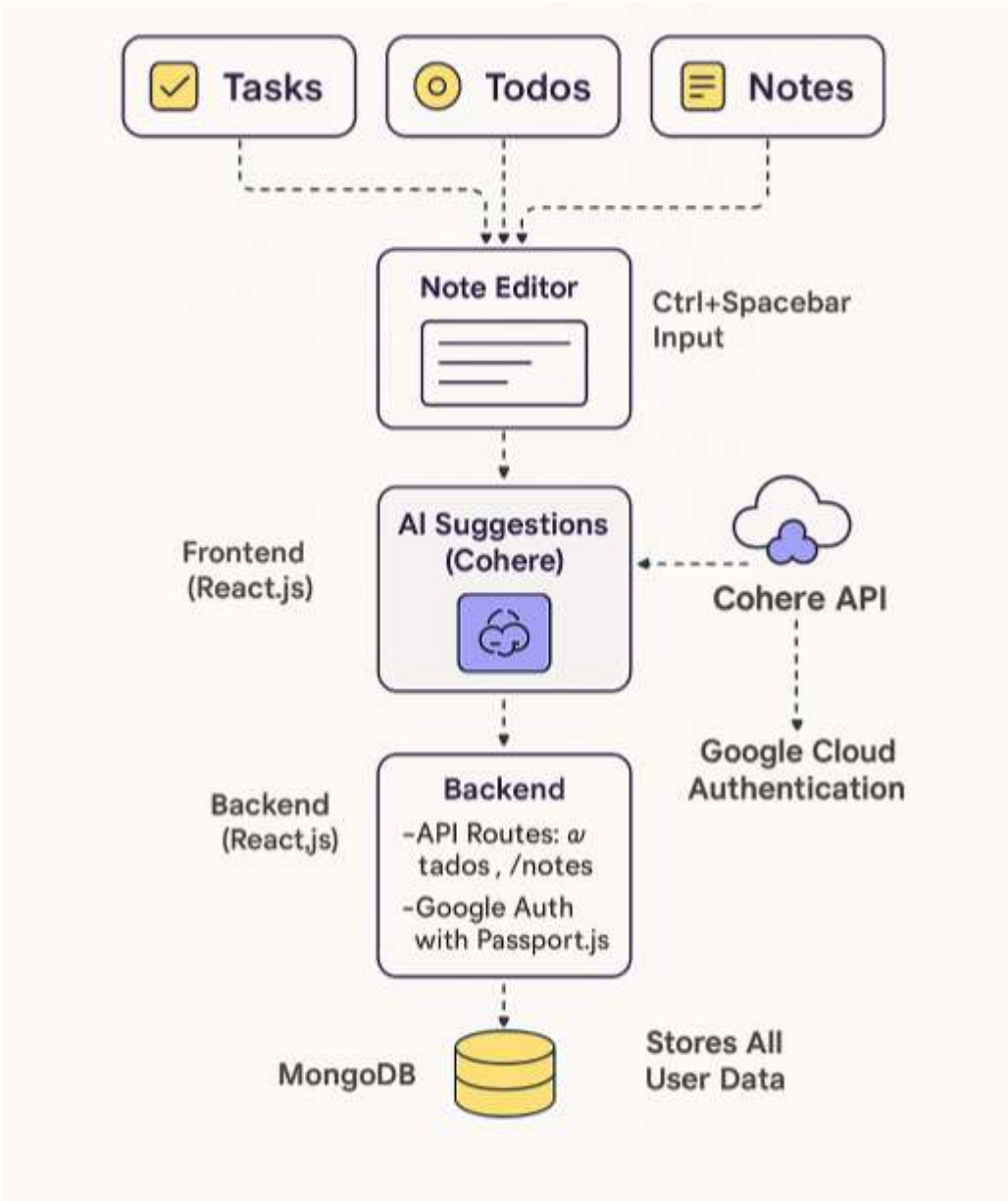

Figure 3.2

3.5. Design selection

The AI-Powered Adaptive Task Management System was selected for implementation due to several key advantages over traditional rule-based task management approaches. Below are the primary factors that led to the selection of this system:

Reasons for Selecting AI-Powered Adaptive Task Management System

1. Higher Efficiency
    - The AI-based system provides real-time, automated prioritization of tasks, eliminating the need for manual categorization.
    - It reduces cognitive load by analyzing user behavior and adjusting task priorities dynamically.
    - The automation of scheduling and organization saves time, making it ideal for individuals managing multiple tasks.
2. User Adaptability
    - Unlike traditional rule-based systems that require users to manually assign priorities, this system learns from user interactions and continuously improves recommendations.
    - The system becomes more accurate and relevant over time as it adapts to personal work habits.
3. Scalability
    - The AI model allows for the integration of additional features in the future, such as predictive scheduling, collaboration tools, and voice recognition.
    - It supports large-scale deployment, making it suitable for personal use as well as organizational and team-based productivity.
4. Competitive Edge
    - The AI-powered approach differentiates Task Buddy from existing solutions that rely on static algorithms.
    - With the integration of machine learning and natural language processing (NLP), the system offers an intuitive and seamless user experience.

Comparison Between Rule-Based and AI-Powered Task Management Systems

Criteria	Rule-Based System	AI-Powered System (Selected)
Task Prioritization	Manual input required	Automated & Adaptive
User Customization	Limited	High customization based on habits
Learning Capability	None	AI-based, learns from user behavior
Scalability	Low	High, can integrate additional AI features
Efficiency	Medium	High, reduces time spent on scheduling
Error Handling	High human error risk	AI reduces manual errors
Task Automation	Minimal	Extensive, AI suggests & adjusts tasks

Table 3.1

3.6. Implementation plan/methodology

To ensure smooth development, a structured implementation methodology will be followed.

Step-by-Step Development Approach

1. Requirement Analysis

- Conduct user surveys to finalize key requirements.
- Identify the technical feasibility of AI-based prioritization.

2. UI/UX Design

- Develop a minimalist, user-friendly front-end using HTML, CSS, and JavaScript.
- Implement drag-and-drop task management for better usability.

3. Backend Development (Node.js & Express.js)

- Design a database schema for storing tasks, user preferences, and AI learning data.
- Develop RESTful APIs for task management operations.

4. AI Integration for Task Prioritization

- Train machine learning models on real-world task management data.
- Implement adaptive learning algorithms to refine prioritization.

5. Collaboration Features Implementation

- Develop task-sharing and team tracking mechanisms.
- Enable real-time updates for collaborative task management.

6. Testing & Debugging

- Conduct unit testing, integration testing, and user acceptance testing.
- Optimize performance and fix bugs before deployment.

7. Deployment & User Feedback

- Launch the application on a cloud server (e.g., AWS, Firebase).
- Collect user feedback and release future updates based on improvements.

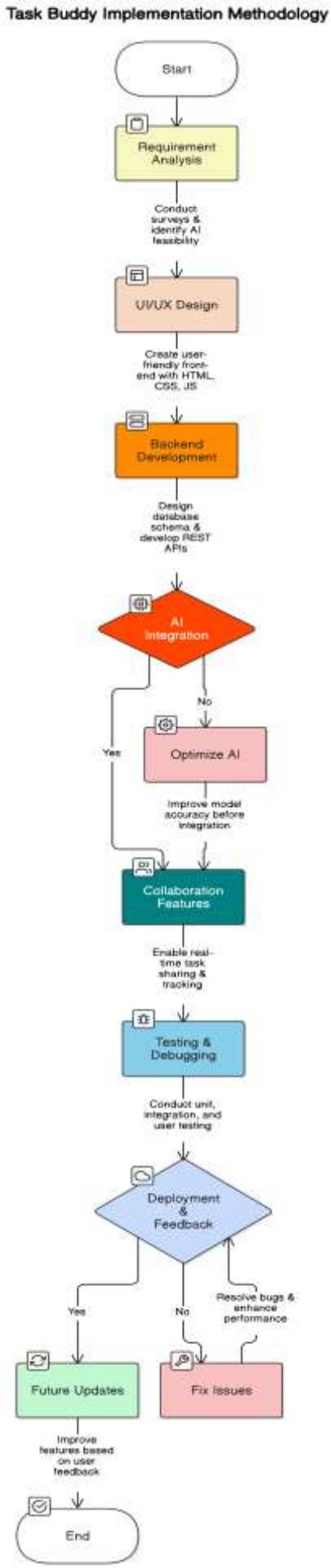
Task Buddy Implementation Methodology
Start
Requirement Analysis
Conduct surveys & identify AI feasibility
UI/UX Design
Create user-friendly front-end with HTML, CSS, JS
Backend Development
Design database schema & develop REST APIs
AI Integration
No
Yes
Optimize AI
Improve model accuracy before integration
Collaboration Features
Enable real-time task sharing & tracking
Testing & Debugging
Conduct unit, integration, and user testing
Deployment & Feedback
Yes
No
Resolve bugs & enhance performance
Future Updates
Fix Issues
Improve features based on user feedback
End

Fig 3.3

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1. Implementation of solution

The implementation of Task Buddy followed a structured approach, integrating modern tools and technologies to ensure a robust and efficient AI-powered task management system. The development process utilized:

- **Analysis**: Surveys and research studies were conducted to understand user needs, analyze existing task management solutions, and identify gaps in the market. AI-driven productivity studies were also referenced to validate the necessity of an adaptive task manager.
- **Design Drawings and Schematics**: Wireframes and UI/UX prototypes were created using tools like Figma to develop an intuitive interface. The backend architecture and AI workflow were designed with scalability in mind.
- **Report Preparation**: Documentation tools such as LaTeX and Microsoft Word were used to prepare structured reports detailing the project's objectives, methodology, and findings.
- **Project Management and Communication**: Agile methodologies and version control systems like GitHub were utilized for project tracking, team collaboration, and iterative development. Regular meetings and discussions ensured alignment with project milestones.
- **Testing, Characterization, and Validation**:
    - **Unit Testing**: Individual modules, including AI-driven prioritization, task scheduling, and reminder functionalities, were tested separately.
    - **Integration Testing**: The communication between the frontend, backend, and AI services was validated to ensure seamless user interactions.
    - **User Testing**: A sample group of users tested the application to provide feedback on usability, responsiveness, and AI-generated task recommendations.
    - **Performance Testing**: The system was tested under different workloads to ensure efficiency and real-time adaptability.

The results indicated that AI-driven prioritization significantly enhanced task management efficiency. However, some users experienced slight delays in AI-based recommendations due to model processing times, suggesting areas for optimization.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1. Conclusion

The Task Buddy project successfully developed an AI-powered to-do list application that provides intelligent task prioritization, adaptive scheduling, and automated reminders. The key outcomes of the project are:

- **Expected Results**:
  - AI-driven task recommendations improved user productivity.
  - Smart scheduling reduced manual workload and decision fatigue.
  - User feedback indicated a preference for AI-assisted task management over traditional to-do list applications.
- **Deviation from Expected Results and Reasons**:
  - Some AI task suggestions were not always perfectly aligned with user preferences due to the initial model requiring more training data.
  - Real-time task adaptation exhibited minor latency, which can be optimized with more efficient algorithms.
  - Limited collaboration features were implemented in the first version, leaving room for future enhancements.

Overall, Task Buddy demonstrated the potential of AI in improving productivity and task management. Despite minor deviations, the system successfully addressed core challenges in prioritization and scheduling.

5.2. Future work

To enhance Task Buddy's effectiveness and user experience, the following modifications and extensions are proposed:

- **Optimization of AI Task Prioritization**: The AI model will be fine-tuned using larger datasets and continuous user interactions to improve accuracy in task recommendations.
- **Reduction in Processing Latency**: Implementing more efficient AI models and optimizing backend processes will help reduce delays in real-time task prioritization.
- **Enhanced Collaboration Features**: Future versions will include team-based task delegation, real-time progress tracking, and shared task lists to support group productivity.
- **Integration with Productivity Tools**: Task Buddy can be integrated with existing productivity applications like Google Calendar, Slack, and Microsoft Teams for seamless workflow management.
- **Predictive Task Scheduling**: AI will be trained to predict future tasks based on past behavior, further automating the scheduling process.
- **Mobile Application Development**: Expanding Task Buddy to Android and iOS platforms will improve accessibility and user engagement.

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APPENDIX

1.Plagiarism Report

TaskBuddywithAI.pdf

ORIGINALITY REPORT

2%	2%	0%	1%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	www.coursehero.com Internet Source	<1%
2	wrap.warwick.ac.uk Internet Source	<1%
3	Submitted to Aspen University Student Paper	<1%
4	Submitted to Colorado Technical University Student Paper	<1%
5	venomswalletfrh.gitbook.io Internet Source	<1%
6	vocal.media Internet Source	<1%
7	www.ijraset.com Internet Source	<1%
8	e-research.siam.edu Internet Source	<1%

2. Design CheckList

The development of the Task Buddy project followed a systematic and structured approach, ensuring that all core components were addressed during the design and implementation phases.

- **User Interface Design**: A responsive and intuitive front-end interface was created using HTML, CSS, and JavaScript to ensure accessibility across devices.
- **Frontend Functionality**: Core features like task creation, editing, deletion, and drag-and-drop management were implemented to enhance user interaction.
- **Backend Development**: Node.js and Express.js were used to handle server-side operations, REST APIs, and communication with the database.
- **Database Design**: MongoDB was used for storing task data, user profiles, and AI-related logs in a structured and scalable format.
- **AI Integration**: An AI module was integrated to provide smart task suggestions and adaptive prioritization based on user behavior.
- **Adaptive Learning**: The system continuously learns from user interactions to improve future recommendations and scheduling.
- **Testing**: Thorough unit and integration testing were conducted to validate individual components and their seamless collaboration.
- **Performance Optimization**: The application performs well under typical usage; minor optimization is ongoing to reduce latency in AI-driven suggestions.
- **Security Measures**: Basic authentication and privacy practices were implemented to safeguard user data.
- **Deployment**: The app was successfully deployed on a cloud platform, ensuring availability and remote access.

This checklist guided the successful implementation of the project and will serve as a foundation for future enhancements.

USER MANUAL

(Complete step by step instructions along with pictures necessary to run the project)

This is a full-stack personal productivity app built with:

- **Frontend**: React
- **Backend**: Node.js + Express
- **Database**: MongoDB (local or Atlas)
- **AI Suggestions**: Cohere API
- **User Authentication**: Google Cloud (email/OTP)

Project Structure

```
project-root/
├── frontend/         # React frontend
│   ├── public/
│   ├── src/
│   │   ├── components/  # Task.js, Todo.js, Note.js, Navbar.js
│   │   ├── pages/       # Home.js, Login.js, Register.js
│   │   ├── App.js, api.js, index.js, App.css
│   └── package.json

└── backend/          # Node.js backend
    ├── Models/
    ├── Routes/
    ├── index.js, passport.js, settings.json
    └── package.json
```

Features

- User login & registration with Google Cloud
- Create, update, delete Tasks, Todos, and Notes
- AI-powered note suggestions via Cohere
- Shortcut: Ctrl + Spacebar for AI suggestion

Prerequisites

- Node.js installed on your machine
- MongoDB (installed locally or hosted on MongoDB Atlas)
- Internet connection for:
    - Google Cloud Authentication
    - Cohere API requests

Setup Instructions

Backend Setup:

1. Open terminal and go to the backend folder: cd backend
2. Install dependencies: npm install
3. Create a file named ".env" inside backend with the following content:

   ```
   PORT=5000
   MONGO_URI=your_mongo_db_uri
   COHERE_API_KEY=your_cohere_api_key
   GOOGLE_CLIENT_ID=your_google_client_id
   GOOGLE_CLIENT_SECRET=your_google_client_secret
   ```

4. Start backend server: node index.js

   Server runs at: http://localhost:5000

Frontend Setup:

1. Go to the frontend folder: cd frontend
2. Install dependencies: npm install
3. Start frontend: npm start

   Frontend runs at: http://localhost:3000

User Authentication:

- Login and registration are handled using Google Cloud credentials.
- Passport.js handles authentication logic.
- settings.json and .env should have correct client ID and secret from Google Cloud Console.

AI Note Suggestions (Cohere):

- While writing a note, press: Ctrl + Spacebar
- This sends your input to the backend.
- The backend sends it to Cohere API.
- Cohere returns a continuation or suggestion.
- The suggestion appears in your note input area.

Example:

Input: Create a note about productivity tips.
AI Response: Start your day with a plan and use time-blocking techniques...

API Routes

Tasks:

- GET /tasks
- POST /tasks
- PUT /tasks/:id
- DELETE /tasks/:id

Todos:

- GET /todos
- POST /todos
- PUT /todos/:id
- DELETE /todos/:id

Notes:

- GET /notes
- POST /notes (can trigger AI)
- PUT /notes/:id
- DELETE /notes/:id

Technologies Used

Feature	Tech
Frontend	React
Backend	Node + Express
Database	MongoDB
AI Suggestion	Cohere API
Auth	Google Cloud
Styling	CSS